

Sri Lanka Institute of Information Technology

Data warehousing and Business

Intelligence

Assignment 2



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Year 03 Semester 01 - WE 01.01

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1.Introduction

This report documents the implementation completed for IT3021 Assignment 2. The work uses the data warehouse that was designed and loaded in Assignment 1 as the source for analytical processing and then

extends the solution with SQL Server Analysis Services (SSAS), Excel-based OLAP analysis, and Power BI reporting.

The implemented solution is based on a car rental domain operating across five branches in Sri Lanka. Three source files were integrated into a SQL Server staging database and then transformed into the CarRental_DW warehouse. That warehouse was then used to build a multidimensional SSAS cube, an Excel workbook connected to the cube, and four Power BI reports.

2. Data source for Assignment 2

The source warehouse is **CarRental_DW** in SQL Server on localhost\SANUJA that was created in Assignment 1. The warehouse stores denormalized dimension data and one central accumulating fact table in a star schema design so that it can support multidimensional analysis and reporting.

2.1 Source systems and staging layer

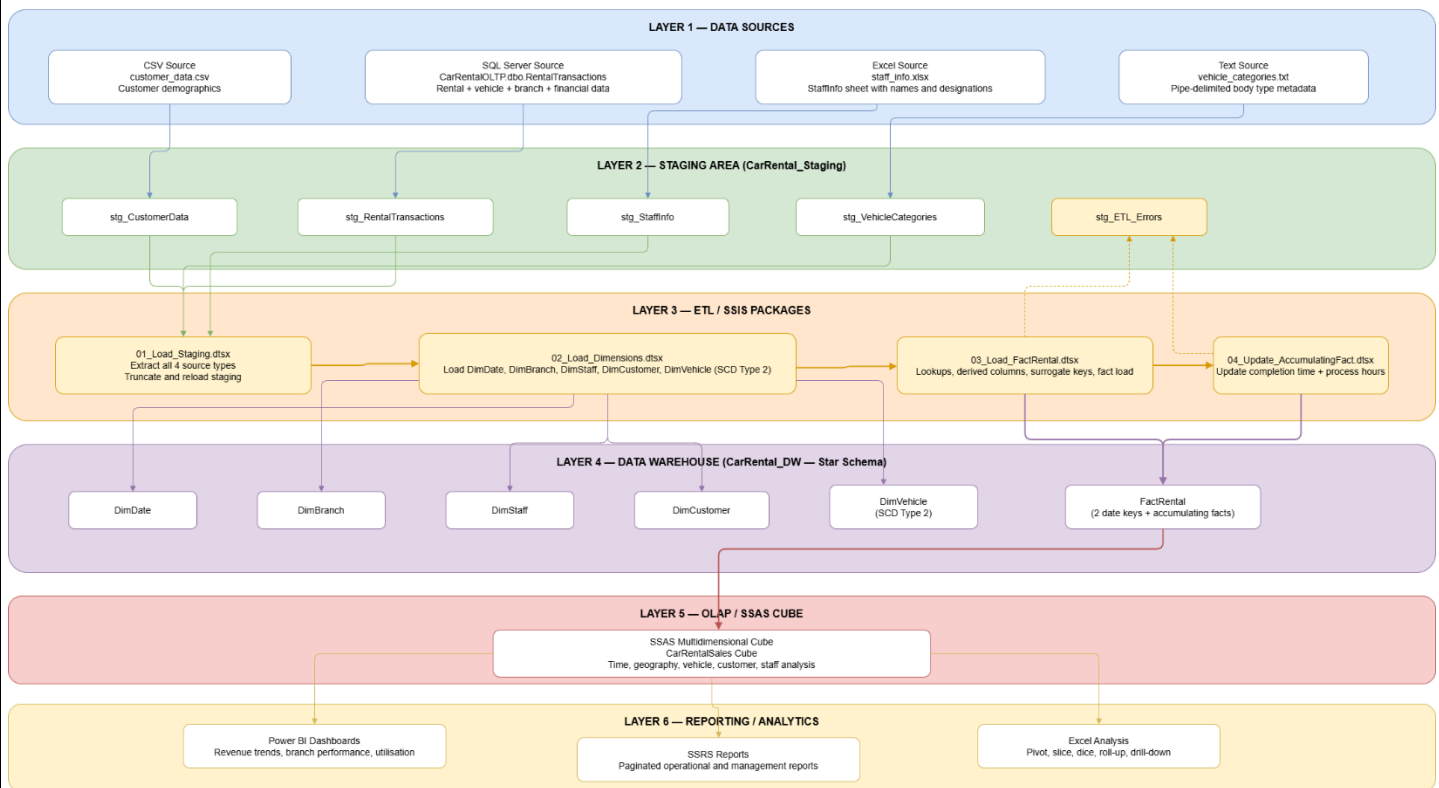
The car rental data integration solution combines three source types. Rental transaction data comes from the flat file rental_data.csv. Customer data comes from the flat file customer_data.csv. Staff master data comes from the Excel file staff_info.xlsx. These files were first extracted into the staging database CarRental_Staging through SSIS and then loaded into the data warehouse.

Source system summary

Layer	Component	Purpose
External source	rental_data.csv	Primary rental transaction source containing 26 columns including dates, vehicle details, branch, payment and cost information. 60,000 records.
External source	customer_data.csv	Customer profile source containing Rental_ID, Customer_Type, Customer_Gender, Customer_Age and License_Type. 60,000 records.
External master data	staff_info.xlsx	Staff master source containing Staff_ID, Staff_Name, Designation, Branch_Name, Email, Phone, Years_Experience and Employment_Type. 14 records.
Staging database	CarRental_Staging	Landing zone used to decouple extraction from warehouse loading and support transformation logic.
Warehouse	CarRental_DW	Star schema warehouse used as the Assignment 2 data source.

Architecture Diagram

Sri Lanka Car Rental DW — Solution Architecture



2.2 Data warehouse design (Retail_DW2)

CarRental_DW was implemented as a star schema. The central fact table is **FactRental**. It is linked to five dimensions: DimDate, DimBranch, DimStaff, DimCustomer and DimVehicle. All analysis is performed through the fact table.

Retail_DW2 warehouse objects

Table	Role in warehouse	Key design notes
DimDate	Date dimension	Stores DateKey, FullDate, DayOfMonth, DayName, DayOfWeek, DayOfYear, WeekOfYear, MonthNumber, MonthName, Quarter, QuarterName, Year and

		IsWeekend. Covers 2022-01-01 to 2023-12-31 (730 days).
DimBranch	Branch dimension	Stores Branch_Name with derived City, Province and Region hierarchy representing 5 Sri Lankan branches.
DimStaff	Staff dimension	Stores Staff_ID, Staff_Name, Designation, Branch_Name, Email, Phone, Years_Experience and Employment_Type for 14 staff members.
DimCustomer	Customer dimension	Stores Customer_Type, Customer_Gender, AgeGroup (derived from Customer_Age) and License_Type representing 48 unique customer segments.
DimVehicle	SCD Type 2 dimension	Stores Car_Brand, Car_Model, Vehicle_Year with history tracking via EffectiveDate, ExpiryDate and IsCurrent. Fuel_Type and Transmission are Type 2 historical attributes.
FactRental	Accumulating fact table	Stores one row per rental transaction with Rental_Duration_Days, Mileage_KM, Daily_Rate_LKR, Rental_Cost_LKR, Insurance_Fee_LKR, Additional_Driver_Fee_LKR, Fuel_Charge_LKR, Total_Amount_LKR and accumulating process timestamps.

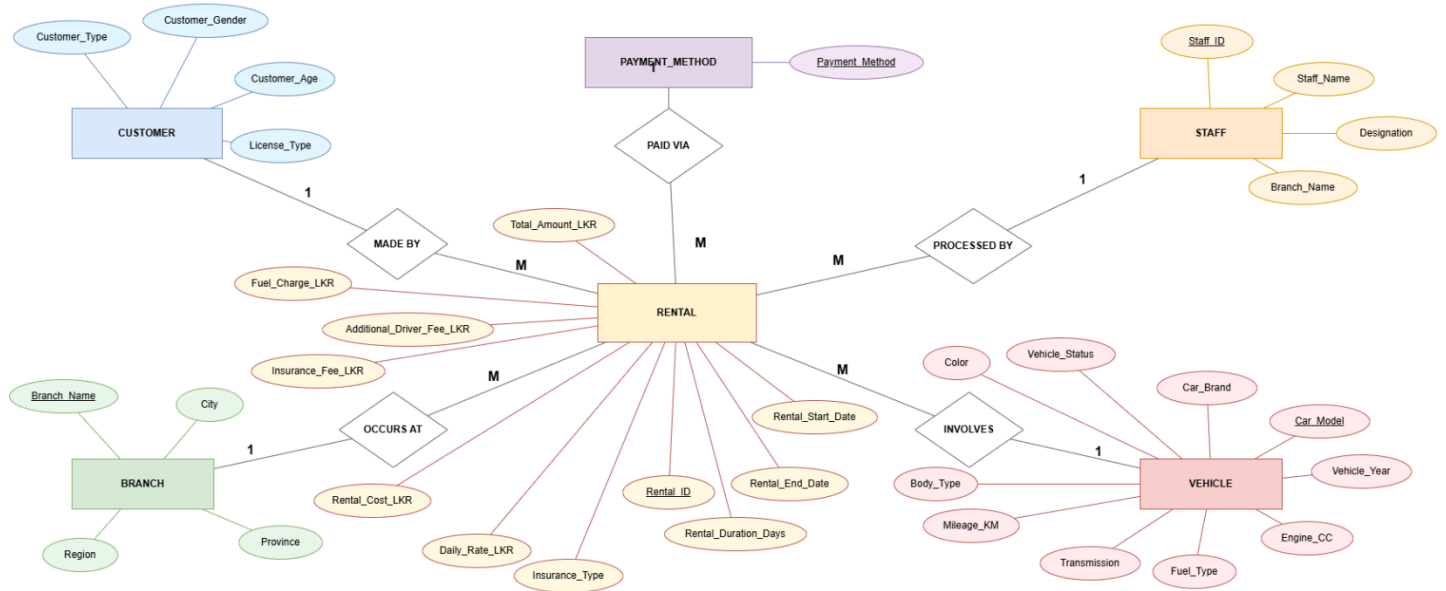
Dimensional Model – Star Schema

CarRental_DW — Star Schema Dimensional Model



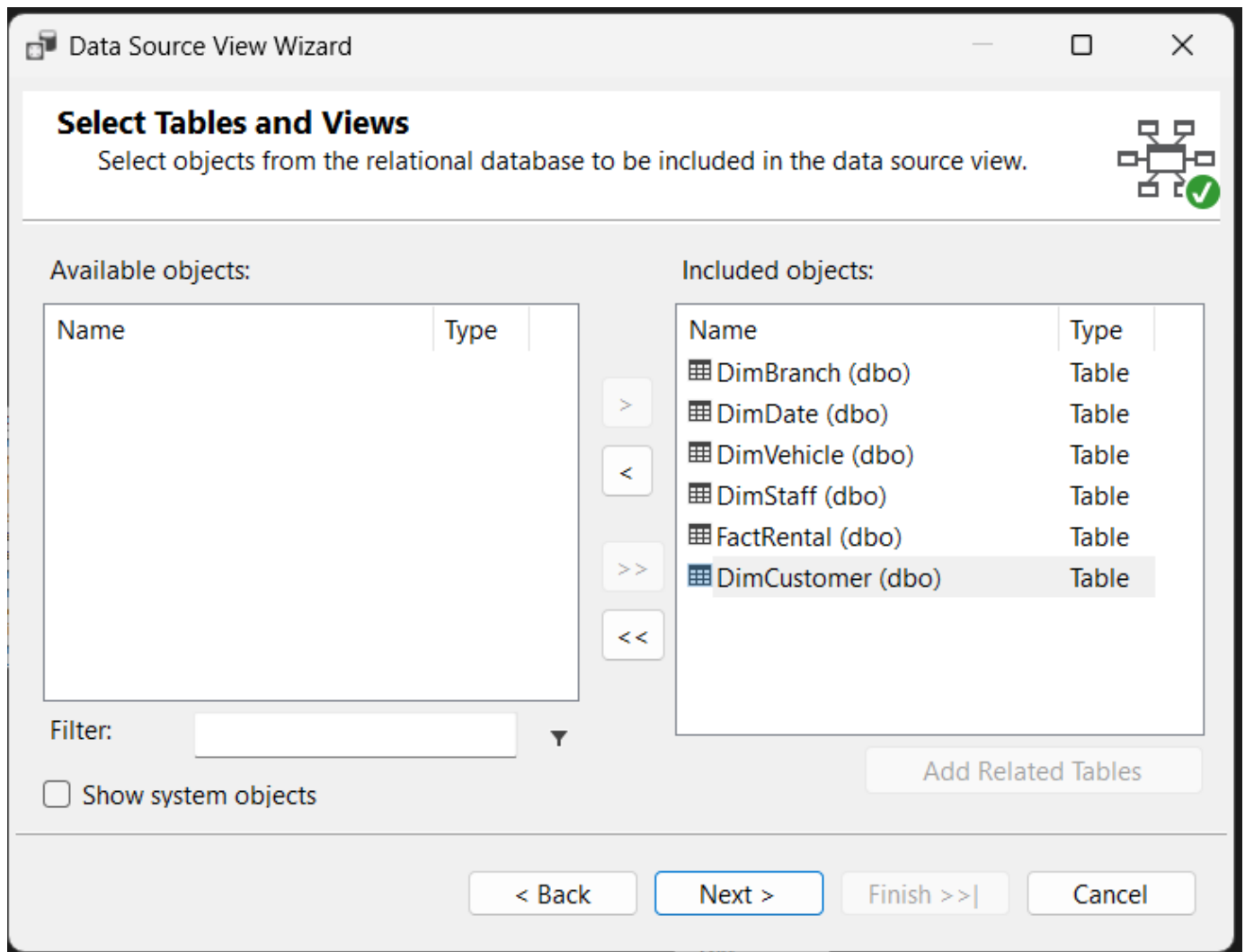
Entity Relationship Diagram

Sri Lanka Car Rental — ER Diagram



Screenshot evidence for Step 1

Data Source View table selection showing the dimension tables and FactRetailSales included for cube modelling.



3. SSAS cube implementation

3.1 Project configuration and deployment targets

The Analysis Services project was created as CarRentalDW_Cube. The relational SQL source used by the data source object was `localhost\SANUJA` with database CarRental_DW. The deployment target for Analysis Services was the multidimensional SSAS instance, and the SSAS database name used during deployment was CarRentalDW_Cube.

SSAS implementation configuration

Project element	Value used
Visual Studio project	CarRentalDW_Cube
SSAS cube name	CarRental_Cube

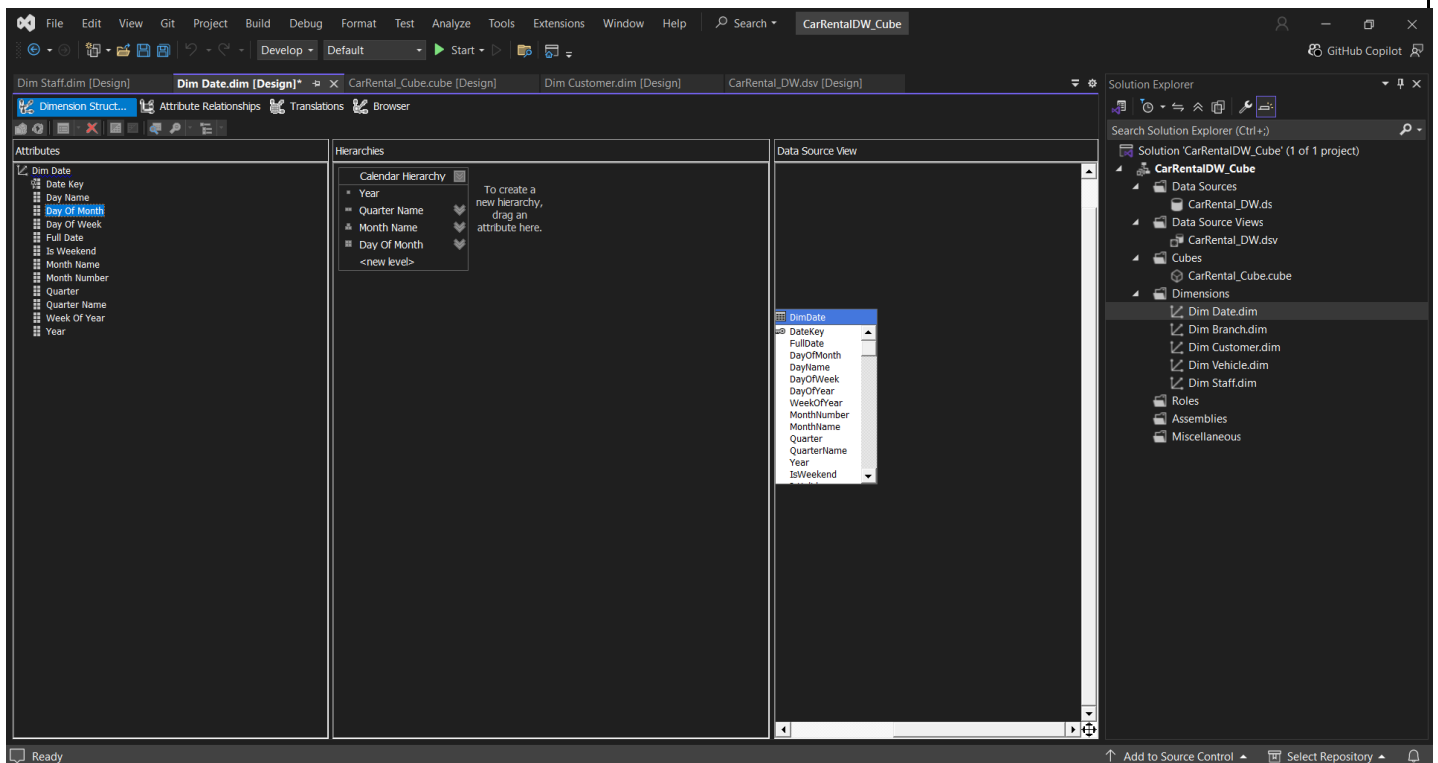
Data source name	CarRental_DW
Data source view	CarRental_DW
Relational SQL source	localhost\SANUJA / CarRental_DW
SSAS server	Multidimensional instance
SSAS database	CarRentalDW_Cube

3.2 Data source, DSV, dimensions and hierarchies

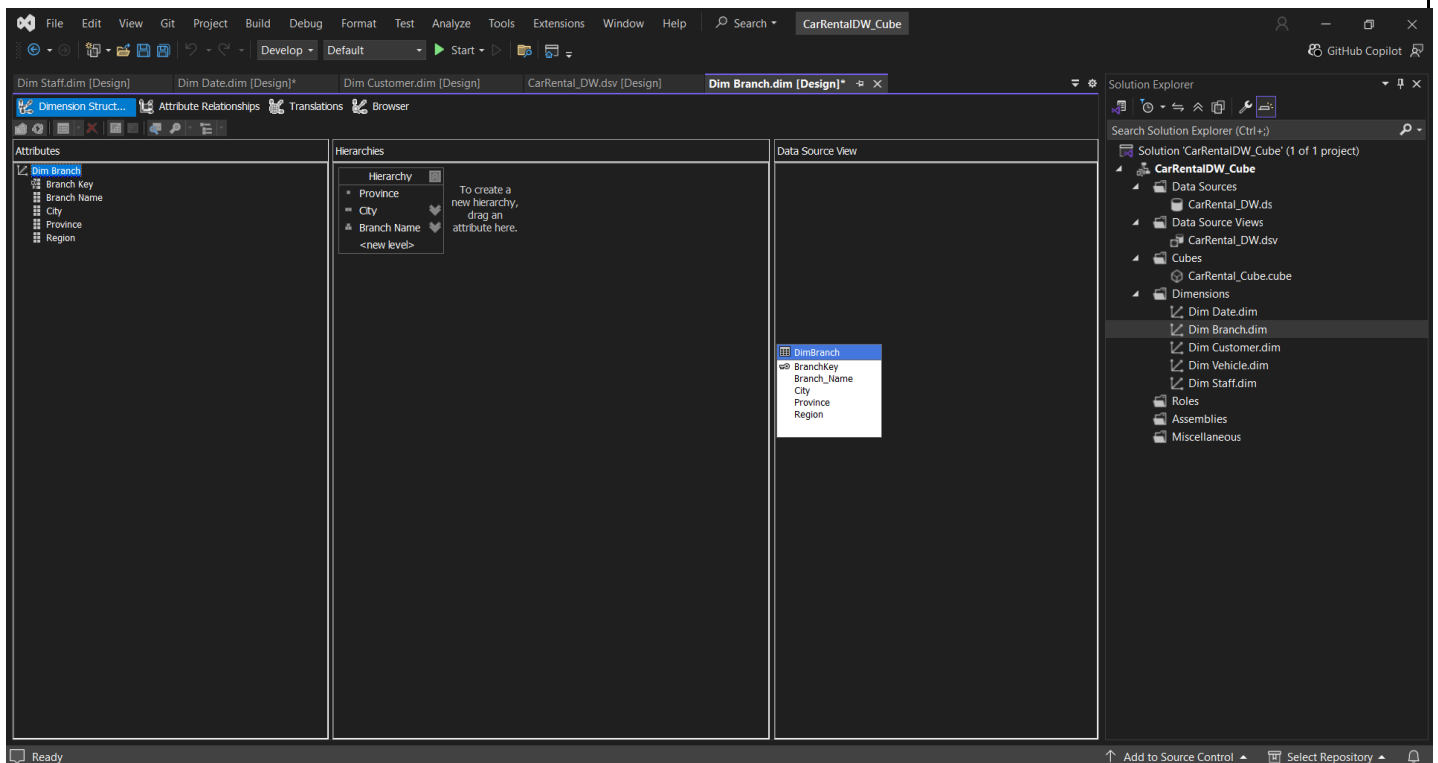
After creating the project, a relational data source was created for CarRental_DW. Then a Data Source View was created using the warehouse tables: DimDate, DimBranch, DimStaff, DimCustomer, DimVehicle and FactRental. Five dimensions were then created based on the dimension tables.

- **Dim Date:** key column DateKey
- **Dim Branch:** key column BranchKey
- **Dim Staff:** key column StaffKey
- **Dim Customer:** key column CustomerKey
- **Dim Vehicle:** key column VehicleKey

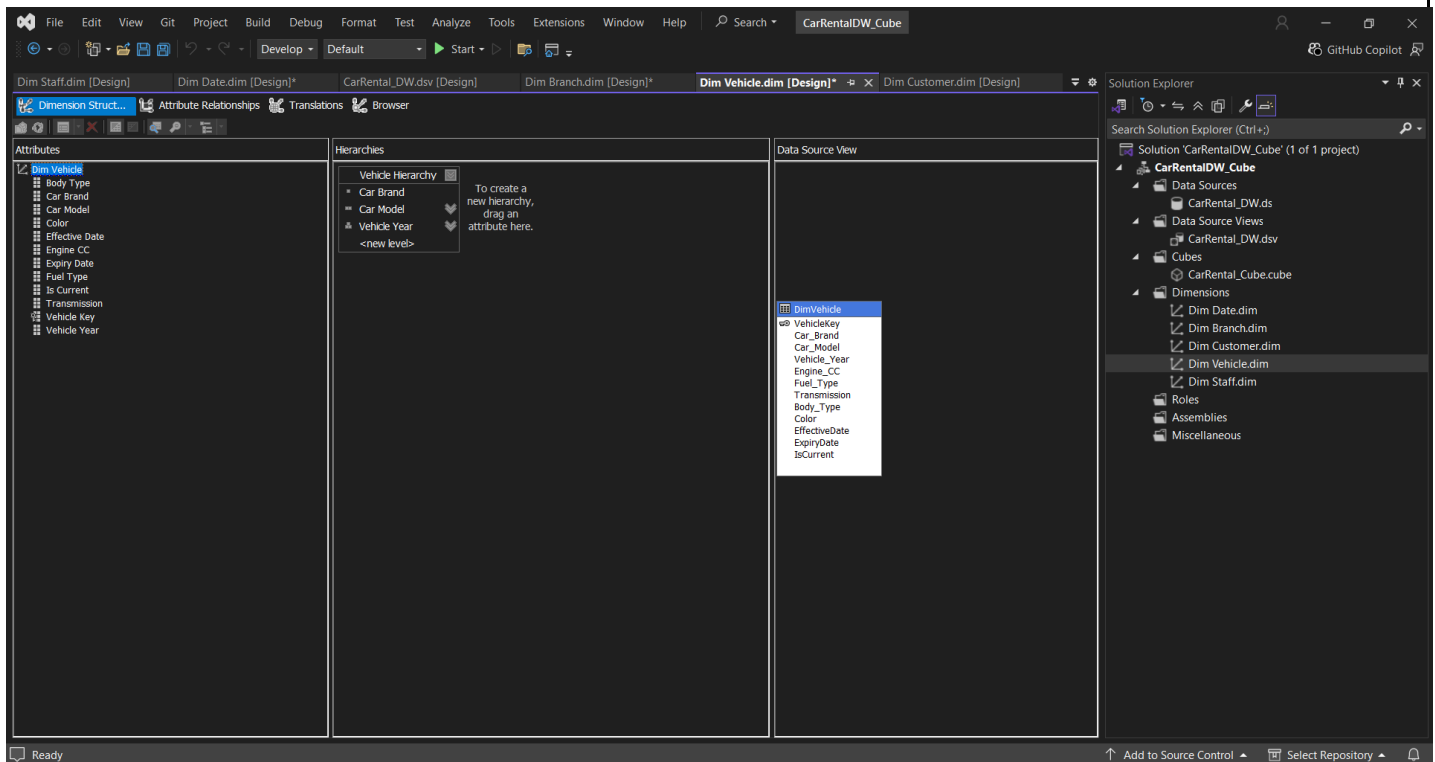
Calendar Hierarchy was implemented in Dim Date. The hierarchy uses Year → Quarter → MonthNumber → DayOfMonth. This was later exposed in cube browsing and also supported drill-down style analysis in the Excel and Power BI stages.



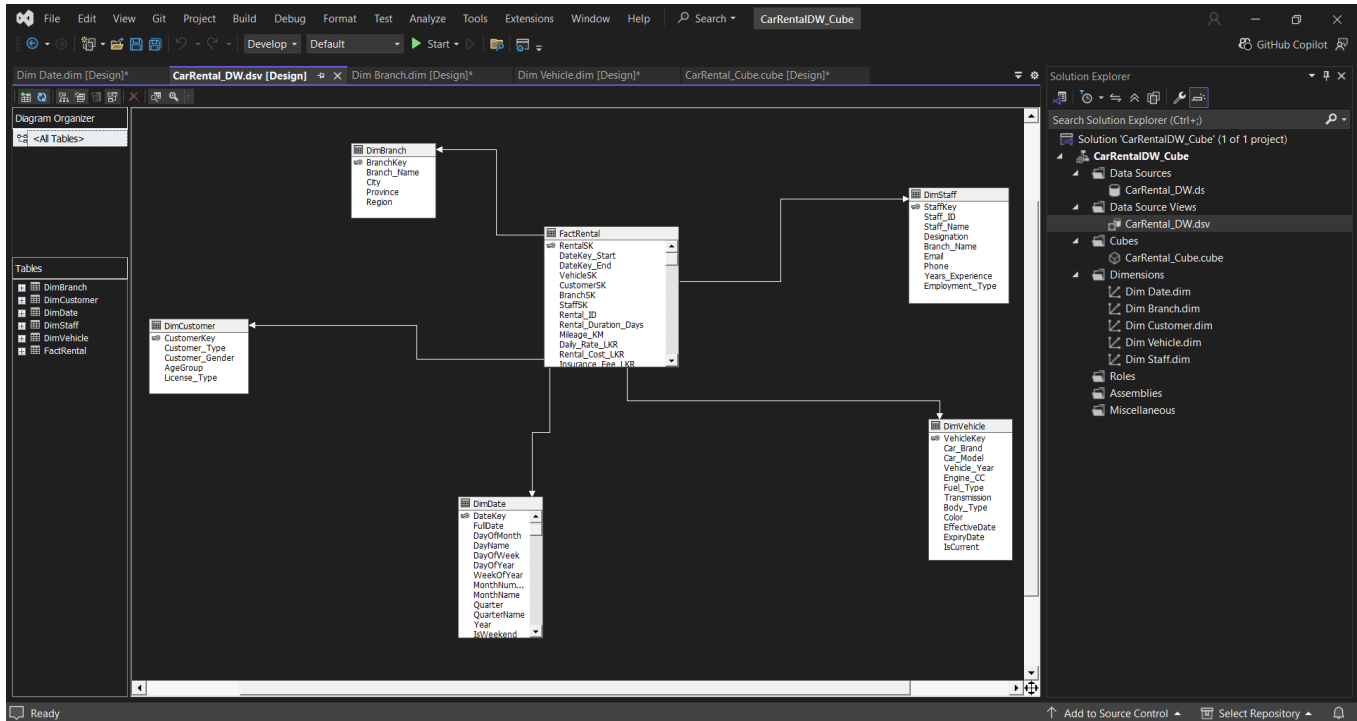
Geography Hierarchy was implemented in Dim Branch. The hierarchy uses Province → City → Branch_Name. This supports geographic roll-up and drill-down analysis across the 5 Sri Lankan branches.



Vehicle Hierarchy was implemented in Dim Vehicle. The hierarchy uses Car_Brand → Car_Model → Vehicle_Year. This supports vehicle-based analysis at brand, model and year levels.



Cube structure and Data Source View after dimensions and cube objects.



3.3 Cube measures and dimension usage

FactRental was selected as the fact table and the measures selected for the cube were Total Amount LKR, Rental Cost LKR, Rental Duration Days, Mileage KM, Insurance Fee LKR, Fuel Charge LKR, Txn Process Time Hours and Fact Rental Count.

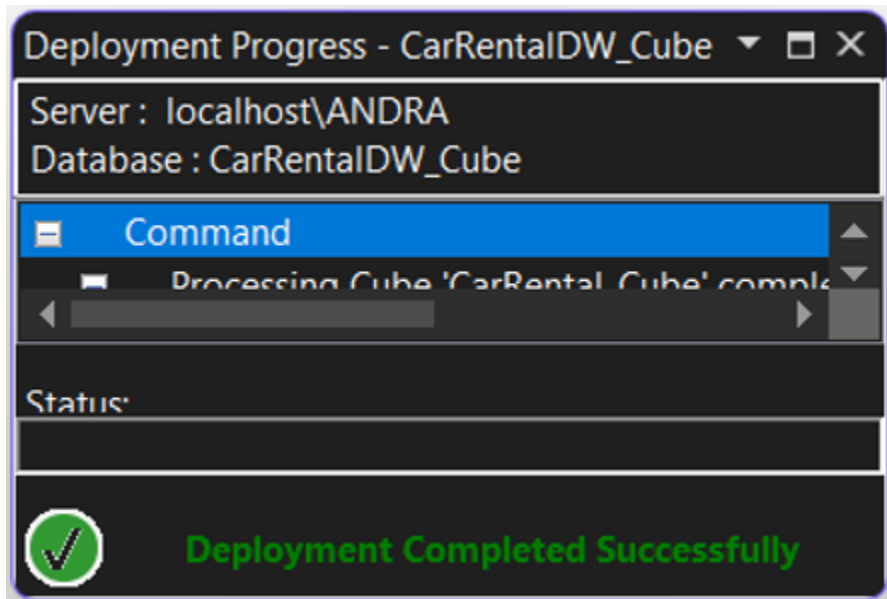
Measures selected in the SSAS cube

Cube measure	Purpose
Total Amount LKR	Total rental revenue analysis
Rental Cost LKR	Base rental cost analysis
Rental Duration Days	Rental length analysis
Mileage KM	Distance travelled analysis
Insurance Fee LKR	Insurance revenue analysis
Fuel Charge LKR	Fuel cost analysis
Txn Process Time Hours	Operational efficiency analysis

Fact Rental Count	Number of rental transactions
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3.4 Deployment, processing and browser validation

Successful deployment of CarRental_Cube to CarRentalDW_Cube



SSAS cube browser showing Total Amount LKR by Branch

Year	Quarter Name	Month Name	Day Of Month	Total Amount LKR
2022	Q1	February	1	15193568.41
2022	Q1	February	10	13370963.1
2022	Q1	February	11	15753062.87
2022	Q1	February	12	17110345.74
2022	Q1	February	13	15420851.76
2022	Q1	February	14	15850261.84
2022	Q1	February	15	14812753.62
2022	Q1	February	16	13706077.33
2022	Q1	February	17	11507948.96
2022	Q1	February	18	17496833.67
2022	Q1	February	19	14475381.59
2022	Q1	February	2	14253862.18
2022	Q1	February	20	14522146.51
2022	Q1	February	21	14037401.8
2022	Q1	February	22	12275332.74
2022	Q1	February	23	12449076.57
2022	Q1	February	24	15582928.04
2022	Q1	February	25	13744931.58
2022	Q1	February	26	12726465.08
2022	Q1	February	27	11216817.5
2022	Q1	February	28	12973609.71
2022	Q1	February	3	14061283.09
2022	Q1	February	4	12793578.91
2022	Q1	February	5	15659593.54
2022	Q1	February	6	14090044.23

SSAS cube browser showing Total Amount LKR by Calendar Hierarchy levels

Dimension	Hierarchy	Operator	Filter Expression	Parameters
Dim Date	<Select hierarchy>			
<Select dimension>				

Year	Quarter Name	Month Name	Day Of Month	Total Amount LKR
2022	Q1	February	1	15193568.41
2022	Q1	February	10	13370963.1
2022	Q1	February	11	15753062.87
2022	Q1	February	12	17110345.74
2022	Q1	February	13	15420851.76
2022	Q1	February	14	15850261.84
2022	Q1	February	15	14812753.62
2022	Q1	February	16	13706077.33
2022	Q1	February	17	11507948.96
2022	Q1	February	18	17496833.67
2022	Q1	February	19	14475381.59
2022	Q1	February	2	14253862.18
2022	Q1	February	20	14522146.51
2022	Q1	February	21	14037401.8
2022	Q1	February	22	12275332.74
2022	Q1	February	23	12449076.57
2022	Q1	February	24	15582928.04
2022	Q1	February	25	13744931.58
2022	Q1	February	26	12736465.08
2022	Q1	February	27	11216817.5
2022	Q1	February	28	12973609.71
2022	Q1	February	3	14061283.09
2022	Q1	February	4	12793578.91
2022	Q1	February	5	15659593.54
2022	Q1	February	6	14090044.23

4. Demonstration of OLAP operations in Excel

4.1 Connecting Excel to the cube

Excel was connected to the deployed SSAS cube using the Data tab. The steps followed were:

1. Open Excel → click **Data** tab
2. Click **Get Data** → **From Database** → **From Analysis Services**
3. Server name: SSAS instance name
4. Select database: CarRentalDW_Cube
5. Select cube: CarRental_Cube
6. Select **PivotTable** → click **OK**

A PivotTable report was created in a new worksheet connected live to the SSAS cube.

4.2 OLAP operation 1 - Roll-up

Roll-up was demonstrated by summarizing Total Amount LKR at the aggregated Year level. This operation reduces the detail level and shows annual totals, moving up the Calendar Hierarchy from daily data to yearly summaries.

Steps followed:

1. Drag Calendar Hierarchy → Year to **Rows**
2. Drag Total Amount LKR to **Values**
3. The pivot shows one row per year — this is roll-up (all daily/monthly data rolled up to year)

The screenshot displays the Microsoft Excel interface with a PivotTable report. The PivotTable is located in the worksheet and shows data for the years 2022 and 2023. The rows are labeled with the year and a sequence number (1, 2, 3, 4). The values are displayed in column E. The Grand Total for 2023 is 7468976404.

Year	Total Amount LKR
2022	
1	1244774562
2	1246925105
3	1244817296
4	1261872946
2023	
1	1228355450
2	1242231044
Grand Total	7468976404

The PivotTable Fields task pane on the right shows the configuration:

- Rows:** Calendar Hierarchy
- Values:** Total Amount LKR

4.3 OLAP operation 2 - Drill-Down

Drill-down was demonstrated by expanding the Calendar Hierarchy from Year to Quarter, then Quarter to Month, then Month to Day. This operation enables the user to start at a summary level and progressively move to lower levels of detail.

Steps followed:

1. Using the same PivotTable from Roll-up
2. Click the + expand button next to **2022**
3. It expands to show Q1, Q2, Q3, Q4 (Quarter level)
4. Click + next to **Q1** to expand to January, February, March (Month level)

The screenshot shows the Microsoft Excel interface with a PivotTable named 'PivotTable1' on the 'Sheet1' tab. The PivotTable is set to show data for the year 2022, expanded to show quarters (Q1, Q2, Q3, Q4), and further expanded to show months. The 'PivotTable Fields' task pane on the right shows the 'Fact Rental' data source with fields like 'Additional Driver Fee LKR', 'Daily Rate LKR', 'Fuel Charge LKR', 'Insurance Fee LKR', and 'Mileage KM'. The 'Calendar Hierarchy' is selected in the 'Rows' area, and 'Total Amount LKR' is selected in the 'Values' area.

Row Labels	Total Amount LKR
2022	
Q1	
February	397266273.2
January	424866018.7
March	422642270
Q2	
April	414512616.6
June	407319885.2
May	425092603.3
Q3	
August	424221957.6
July	418423330
September	402172008.1
Q4	
December	424918207.7
November	415040227.7
October	421914511
2023	2470586494
Grand Total	7468976404

4.4 OLAP operation 3 – Slice

Slice was demonstrated by applying a filter on a single dimension value. This produces a single dimensional cut of the cube and allows focused analysis of a specific subset of the data.

Steps followed:

1. Create a new PivotTable
2. Drag Calendar Hierarchy to **Rows**
3. Drag Total Amount LKR to **Values**
4. Drag Branch_Name to **Filters** area (top)
5. Click the filter dropdown → select only **Negombo Airport Rentals**
6. The pivot now shows **ONLY** Negombo data across all years and months

The screenshot shows an Excel PivotTable with the following data:

Row Labels	Total Amount LKR
2022	
Q1	
February	81677525.5
January	82438170.52
March	83871105.07
Q2	
April	82973667.48
June	84726990.86
May	93821393.6
Q3	
August	88227404.11
July	88451861.58
September	79783898.75
Q4	
December	89346083.59
November	88043689.8
October	82731063.75
2023	505955250.2
Grand Total	1532048105

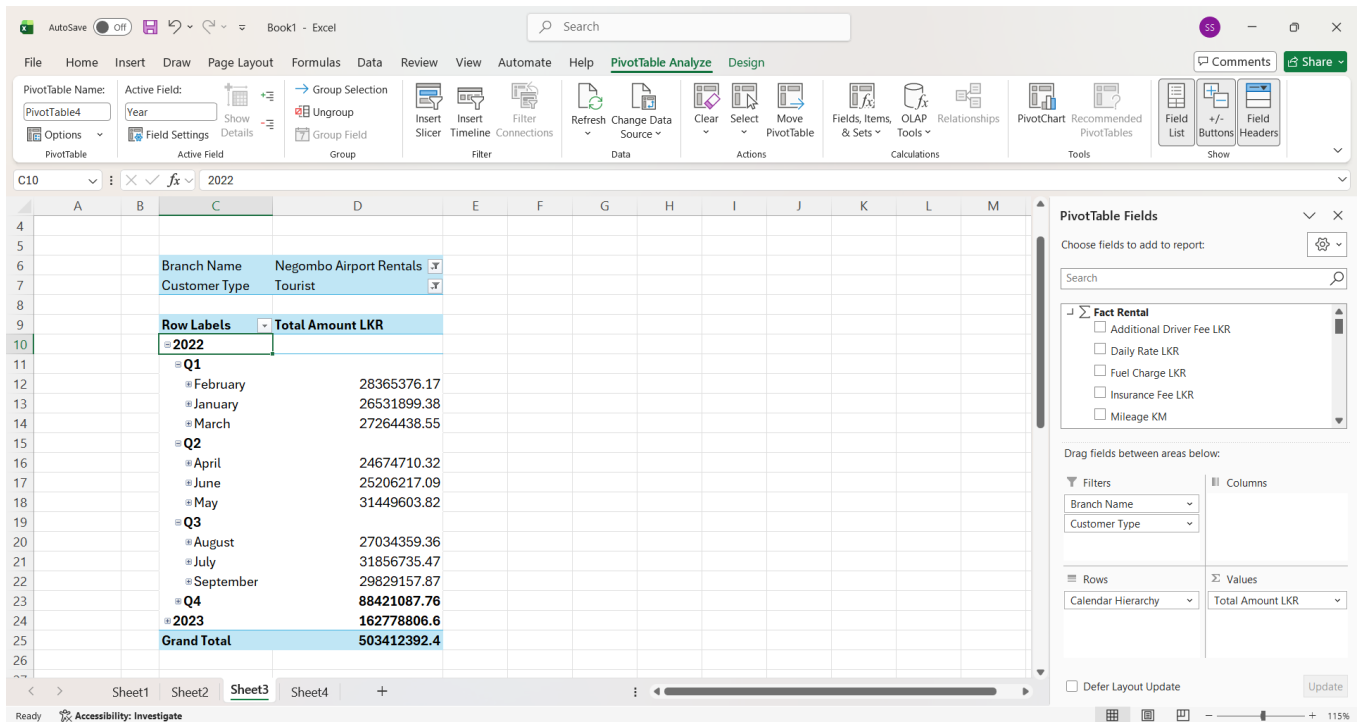
4.5 OLAP operation 4 – Dice

Dice was demonstrated by filtering multiple dimensions simultaneously. This creates a smaller sub-cube for detailed analysis under multiple conditions.

Steps followed:

1. Create a new PivotTable
2. Drag Calendar Hierarchy to **Rows**
3. Drag Total Amount LKR to **Values**
4. Drag Province to **Filters** → select **Western** only
5. Drag Customer_Type to **Filters** → select **Tourist** only

6. Both filters applied simultaneously — this is the dice operation (two-dimensional cut)



4.6 OLAP operation 5 – Pivot

Pivot was demonstrated by rotating the analytical view by moving Car_Brand and Province between Rows and Columns. This preserves the same measure values but presents them from another analytical angle, making it easier to compare by vehicle brand or by province.

Steps followed:

Before pivot:

1. Create new PivotTable
2. Drag Car_Brand to **Rows**
3. Drag Province to **Columns**
4. Drag Total Amount LKR to **Values**

After pivot (rotate the axes): 5. Drag Province from Columns → move to **Rows** 6. Drag Car_Brand from Rows → move to **Columns** 7. Now Province is in rows and Car_Brand is in columns — axes swapped

Excel interface showing a PivotTable and the PivotTable Fields task pane.

PivotTable:

Row Labels	Central	Northern	Southern	Western	Grand Total
Honda	257811120.1	241820598.5	254193131.9	498827880.6	1252652731
Hyundai	255839873.8	234836604.8	249813958.2	500318619.8	1240809057
Mitsubishi	250661715.5	243912456.8	249633754.7	483632615.5	1227840542
Nissan	251513214.5	237425131.6	256473594	504098269	1249510209
Suzuki	248805484.8	252876720	240731853.8	498560211.8	1240974270
Toyota	253877552.5	241024969.7	250989772.3	511297299.6	1257189594
Grand Total	1518508961	1451896481	1501836065	2996734896	7468976404

PivotTable Fields:

- Fact Rental
 - ☐ Additional Driver Fee LKR
 - ☐ Daily Rate LKR
 - ☐ Fuel Charge LKR
 - ☐ Insurance Fee LKR
 - ☐ Mileage KM

Drag fields between areas below:

Filters	Columns
	Province

Rows	Values
Car Brand	Total Amount LKR

☐ Defer Layout Update Update

5. Power BI Reports

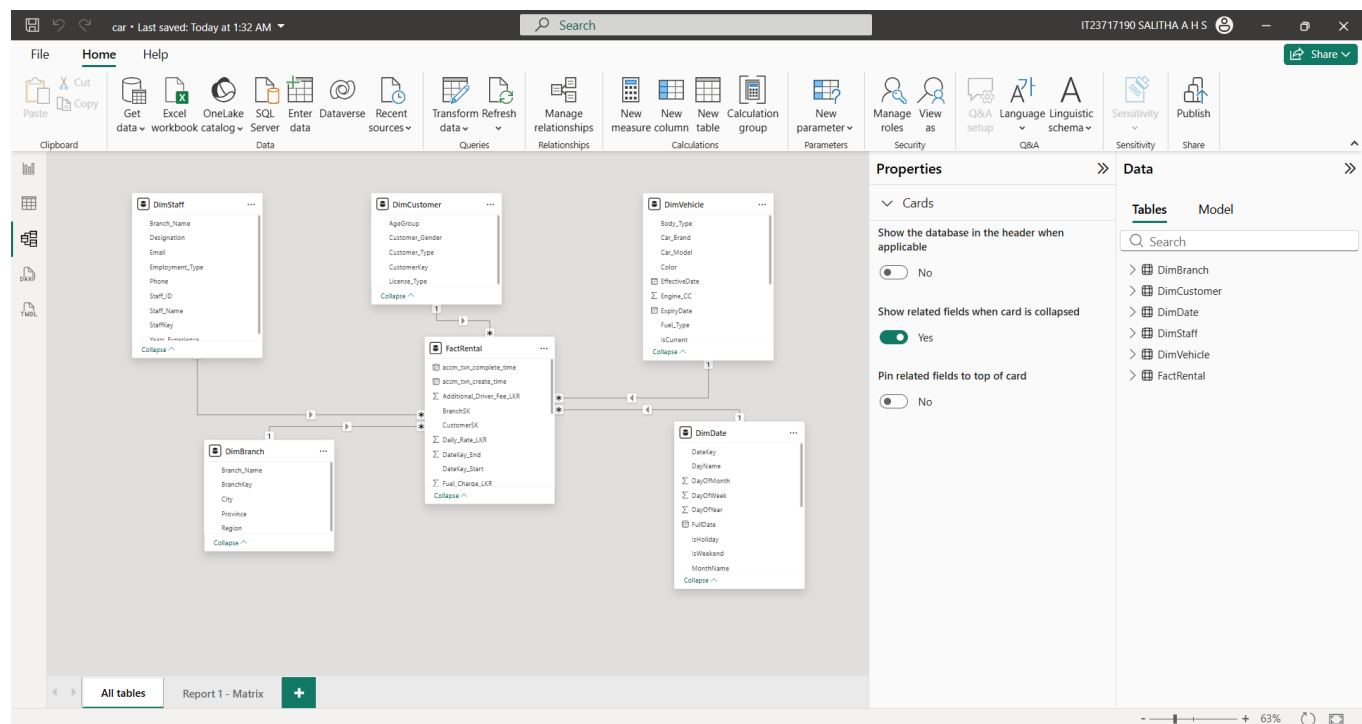
5.1 Data preparation and model relationships

Power BI imported DimDate, DimBranch, DimStaff, DimCustomer, DimVehicle and FactRental from `localhost\SANUJA / CarRental_DW`. The model view confirmed the correct star-schema relationships, with the dimension tables on the 1-side and FactRental on the many-side.

Relationships configured in Power BI Model view:

From table	From column	To table	To column	Cardinality
FactRental	DateKey_Start	DimDate	DateKey	Many to One
FactRental	BranchSK	DimBranch	BranchKey	Many to One
FactRental	VehicleSK	DimVehicle	VehicleKey	Many to One
FactRental	CustomerSK	DimCustomer	CustomerKey	Many to One
FactRental	StaffSK	DimStaff	StaffKey	Many to One

A Date Hierarchy was also created in Power BI on the DimDate table with levels: Year → Quarter → MonthName → DayOfMonth.

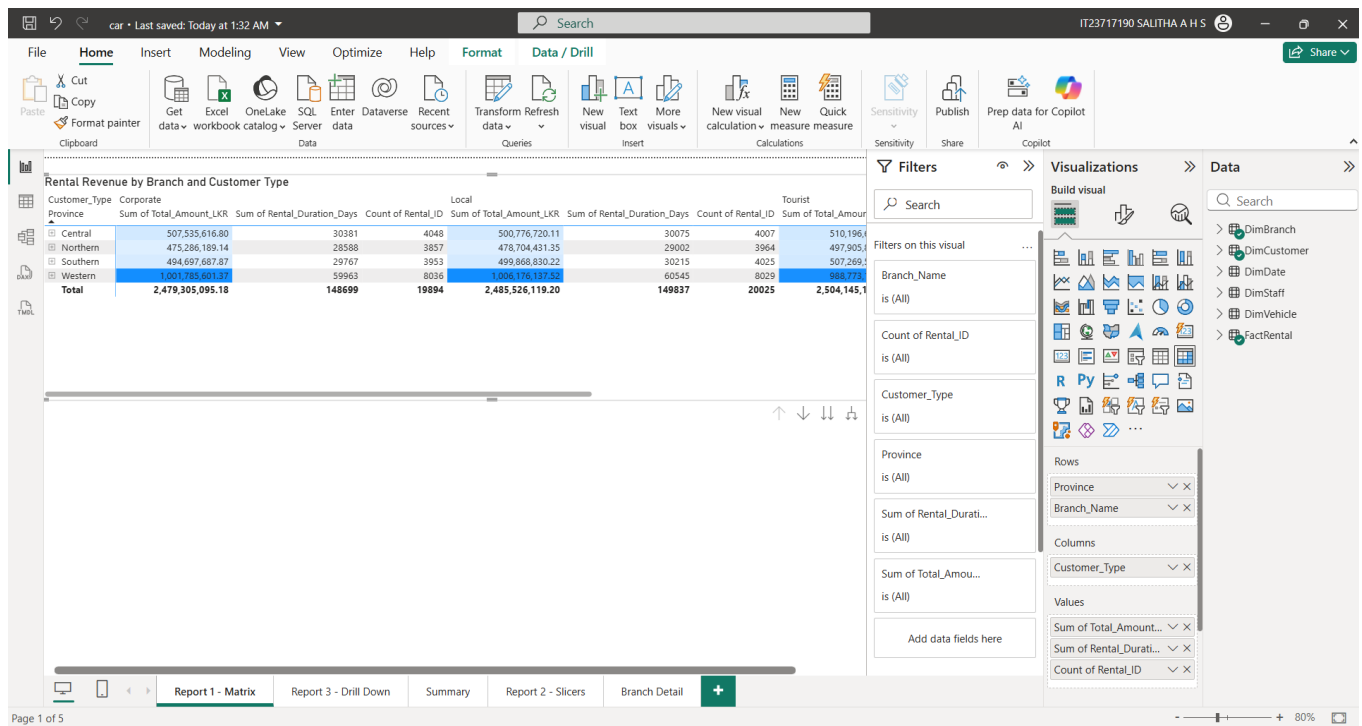


5.2 Report 1 - Matrix visual

Report 1 used a Matrix visual to display detailed tabular data with row and column groupings. Province and Branch_Name were placed on rows to show the geographic hierarchy. Customer_Type was placed on columns to compare customer segments. The measures Total_Amount_LKR (Sum), Rental_Duration_Days (Average) and Rental_ID (Count) from FactRental were added to the Values section.

Configuration:

Area	Field
Rows	DimBranch → Province, Branch_Name
Columns	DimCustomer → Customer_Type
Values	FactRental → Total_Amount_LKR (Sum)
Values	FactRental → Rental_Duration_Days (Average)
Values	FactRental → Rental_ID (Count)

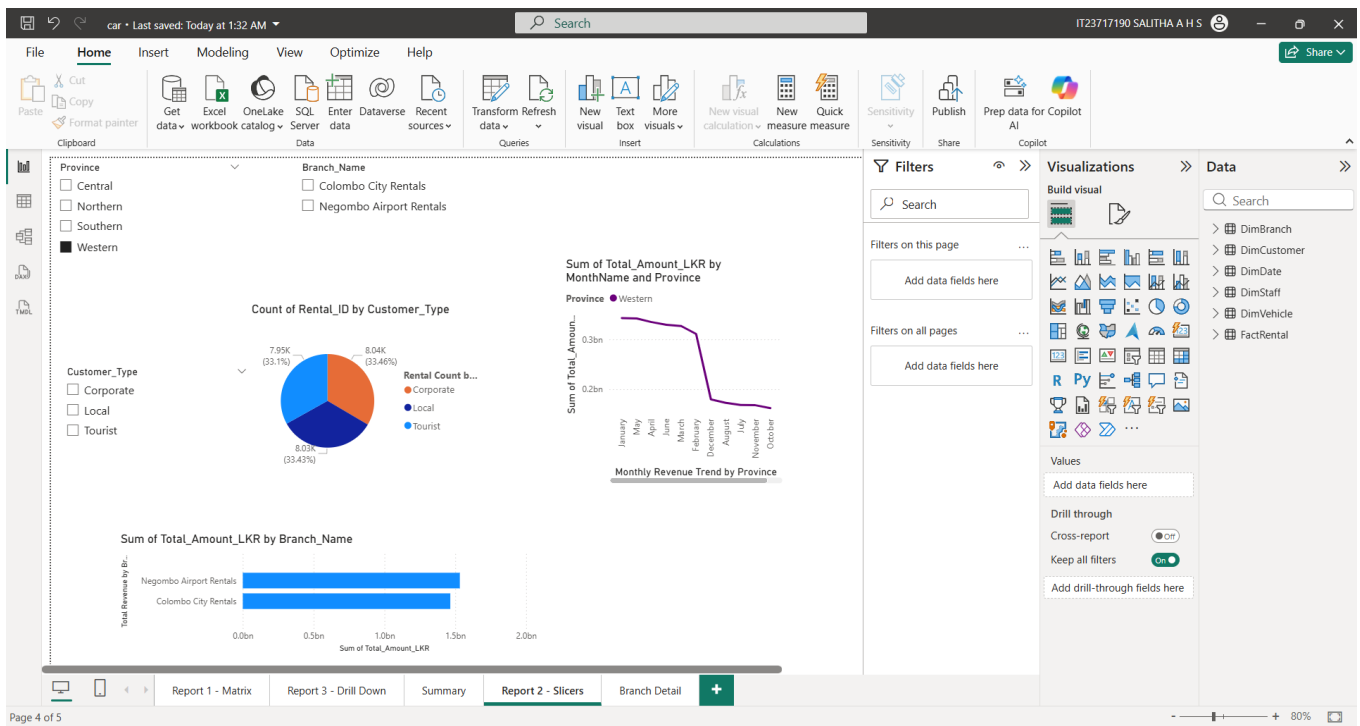


5.3 Report 2 - Multiple slicers with cascading filter

Report 2 was developed as a slicer-driven analysis page using cascading geographic filters. The first slicer is **Province**, and the second slicer is **Branch_Name**. When a province is selected, the Branch_Name slicer is automatically restricted to only the branches belonging to that province. For example, selecting Western Province shows only Negombo Airport Rentals and Colombo City Rentals. The page contains four visuals:

1. **Column chart** was created using MonthName from DimDate and Sum of Total_Amount_LKR from FactRental to show how rental revenue changes across months.
2. **Pie chart** was created using Customer_Type from DimCustomer and Sum of Total_Amount_LKR to show the contribution of each customer type to total revenue.
3. **Horizontal bar chart** was created using Car_Brand from DimVehicle and Sum of Rental_Duration_Days to compare total rental days per vehicle brand.
4. **Card visual** displaying the overall Sum of Total_Amount_LKR.

This report allows users to narrow analysis by geographic location while simultaneously viewing revenue trends over time, customer type breakdown, and vehicle brand performance.



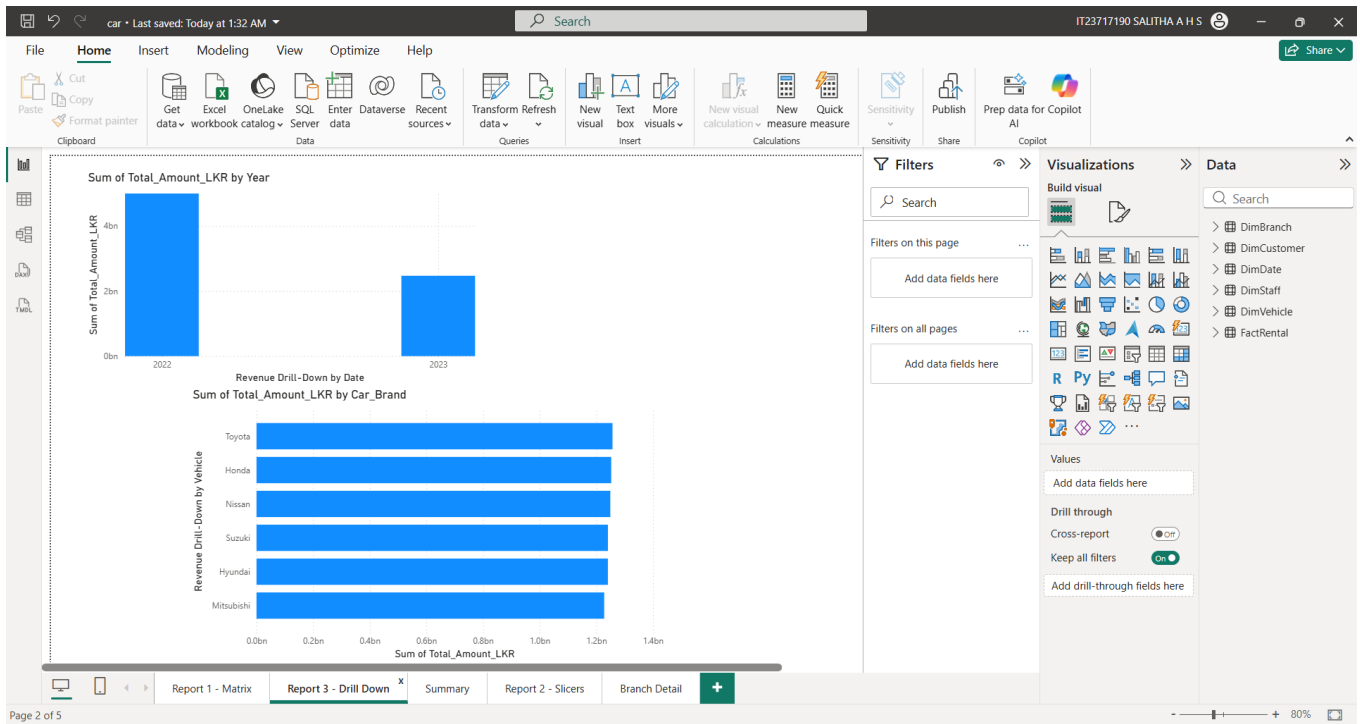
5.4 Report 3 – Drill-down report

This report was created for drill-down analysis using the date hierarchy. A clustered column chart was inserted, and fields from DimDate were added in hierarchical order: Year → Quarter → MonthName → DayOfMonth. Total_Amount_LKR was used as the value field with sum aggregation.

The drill-down feature was enabled by clicking the drill-down arrow icon in the chart header. Users can click on any bar to drill from Year level to Quarter level, then from Quarter to Month, and from Month to Day level.

Steps followed:

1. Insert Clustered Column Chart
2. X-axis: DimDate → Year, Quarter, MonthName, DayOfMonth (in order)
3. Y-axis: FactRental → Total_Amount_LKR
4. Enable drill-down mode using the ↓↓ icon
5. Click 2022 bar → drills to Q1, Q2, Q3, Q4
6. Click Q1 → drills to January, February, March



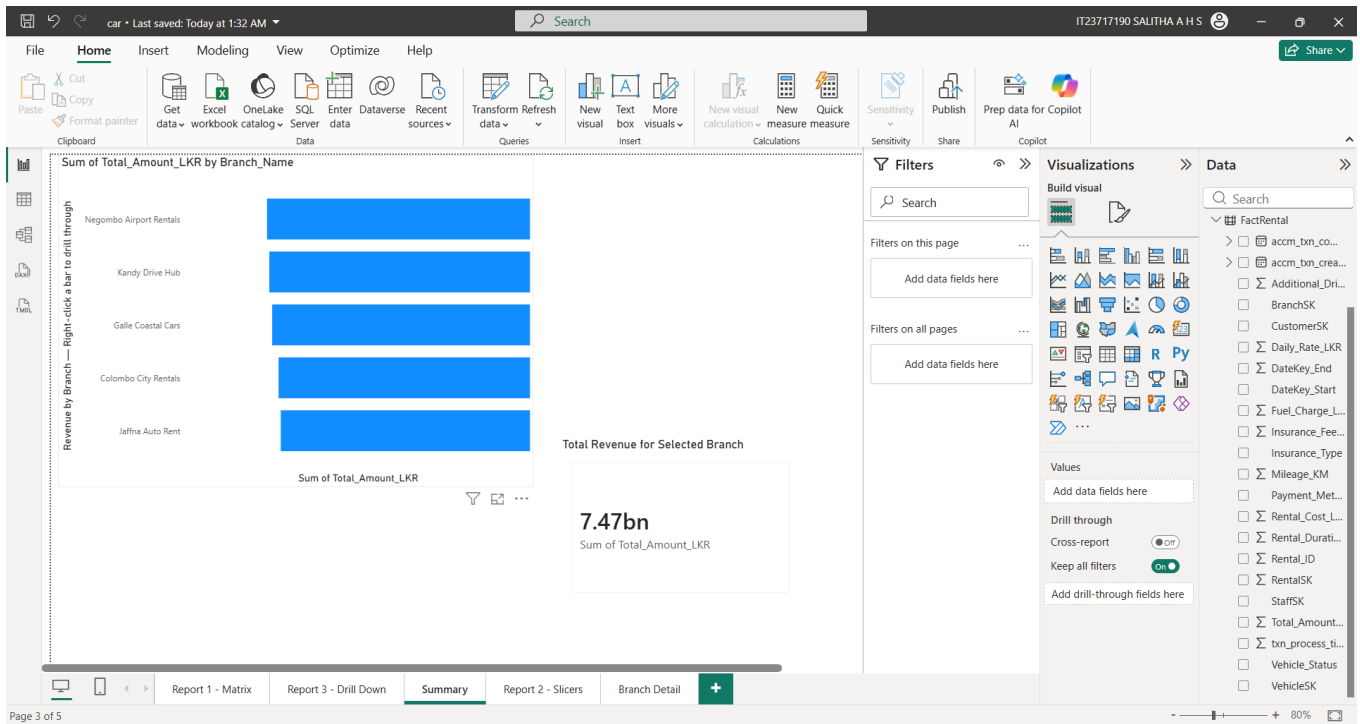
5.5 Report 4 – Drill-through report

This was designed for drill-through functionality. A summary page was created first showing Total Amount LKR by Branch_Name in a bar chart. A dedicated detail page was then created. On the detail page, **Branch_Name** from DimBranch was added to the Drill-through filters area. When a user right-clicks a branch bar on the summary page and selects drill-through, Power BI opens the detail page and automatically filters all visuals to the selected branch.

The drill-through detail page contains three visuals. A **table visual** was used to show detailed rental records for the selected branch including Staff_Name, Car_Brand, Car_Model, Total_Amount_LKR, Rental_Duration_Days and FullDate. A **card visual** was used to display the total revenue summary for the selected branch. A **bar chart** was used to show revenue by vehicle brand for the selected branch.

Steps followed:

1. Create Summary page → add bar chart (X: Total_Amount_LKR, Y: Branch_Name)
2. Create new page → rename to Branch Detail
3. In Filters pane → Drill through section → drag Branch_Name
4. A back button appears automatically
5. Add Table visual with rental detail columns
6. Add Card visual with Total_Amount_LKR
7. Add Bar Chart with Car_Brand vs Total_Amount_LKR
8. Test: go to Summary → right-click Negombo bar → Drill through → Branch Detail



5.6 Publishing to Power BI Service

After all four report pages were created and tested in Power BI Desktop, the report file was published to Power BI Service using the Publish option in the Home tab.

Steps followed:

1. Click **Home** tab → **Publish**
2. Select workspace: **My Workspace**
3. Click **Select**
4. Wait for publishing to complete

